

Chapter 12

Family of exclusives in Polish

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I define the semantics of four exclusive adverbials in Polish – *tylko*, *zaledwie*, *dopiero*, and *aż* – as composed of three core components: an assertion of exclusivity, a factive presupposition and a scalar presupposition. By minimally adjusting parameters such as the direction and dimension of the scale and the size of the associated focus, different flavors of exclusivity emerge. These parameter settings give rise to temporal readings (*dopiero*), approximative and counterfactual readings (*zaledwie*), and even additive-like readings (*aż*).

1 Introduction

This chapter presents the behavior of four exclusive adverbials in Polish, *tylko*, *zaledwie*, *dopiero* and *aż*, and shows how they can be defined as a family: Each lexical entry includes a common core (assertion of exclusivity, presupposition of scalarity) with minimal differences in specific parameters (the direction and the dimension of the scale, the size of the associated focus).¹ We will start with a context where all four adverbs have a similar effect on the meaning and then use a battery of tests to identify the individual meaning components.

All of the adverbs contribute to the sentence in (1) the additional meaning that the 30th place is a weak result, i.e. the team did not perform any better than that.

¹The glosses of the Polish particles in the examples reflect the parallels with the English counterparts. In Section 4, we will see that *tylko* is similar to *only* and *zaledwie* to *merely*. *Dopiero* has a German counterpart, *erst*, which has been named *TEMPORAL only*. *Aż* on my view is a scalar opposite of *merely*, so it has no approximate translation and is glossed here as *AZ*.



- (1) Zespół był słaby i zajął {tylko / zaledwie / dopiero / aż} 30
team was poor and achieved only merely only_{TEMP} AŻ 30th
miejsce.
place
'The team performed poorly and earned only 30th place.'

English *only* has exactly this meaning '(at least) X and nothing more', with 'nothing more' excluding all contextually ranked higher alternatives. Is it only a coincidence that in this particular context the four adverbials seem to mean 'only'? Could the sentence rather be saying that 'The team was ranked as low as the 30th place'? This meaning is similar but it does not contain the exclusive component 'nothing more'. I argue that the four adverbial particles should all be classified as exclusives, i.e., they all share the exclusive meaning component, and that this primary component is at-issue (Potts 2005, Beaver & Clark 2008, Simons et al. 2011, Roberts 2011). Each one of them, however, has its own flavor so that in most contexts they are not interchangeable as they are in (1). I identify the differences in their use that make them similar only under certain conditions. It turns out that Polish does not have one adverbial that can contribute the exclusive meaning in all contexts, the way English *only* can. Instead, each of the items places particular restrictions on the dimension of scale of the excluded alternatives. Furthermore, the direction of exclusion differs: alternatives both higher and lower on the scale can be excluded (Tomaszewicz 2012, Tomaszewicz 2013a,b). I propose lexical entries for each exclusive such that they all include a common core with minimal differences in specific parameters.

2 Exclusivity and scalarity

The adverb *only* contributes the meaning of exclusion: in contrast to (2), the sentence in (3) states that Maria met Jan and she did not meet anyone else.

- (2) Maria met Jan.
(3) Maria met only Jan.

In some contexts *only* additionally contributes scalarity by implying a ranking among the excluded alternatives. Consider how in (4) we understand that the 30th place is not a good result, even though we do not know how many places were assigned in total. In (5) it is implied that 9 PM is late, and in (6)–(7) that the manager is lower on the scale than some contextually relevant alternative.

- (4) Jan got only the 30th place.
- (5) Jan arrived only at 9 PM.
- (6) Jan is only/merely a manager.
- (7) Maria talked only/merely to the manager.

In the example in (4) it is obvious that you can be ranked at a single place only, so the exclusive meaning should be paraphrased as ‘Jan got no better place than the 30th’ rather than ‘Jan got the 30th place and no other place’ in contrast to the paraphrase of the sentence in (3). These two readings of *only* have been called SCALAR and QUANTIFICATIONAL (Bonomi & Casalegno 1993, Krifka 1993, Klinedinst 2005, Riester 2006, Beaver & Clark 2008, Zeevat 2009, among others). The scalar reading of the sentence in (5) is roughly equivalent to ‘Jan arrived no sooner than 9 PM’ because the most plausible scenario is that Jan arrived once, as opposed to ‘Jan arrived at 9 PM and at no other time’, which would be the quantificational reading. In contrast to (4), in (6) Jan is a manager in addition to being other obvious things such as human, terrestrial, mortal... These properties are not ordered on the scale with respect to the property of being a manager, they are not contextually relevant. Thus, in (6), *only* evokes a contextually relevant scale and places ‘manager’ low on that scale, as illustrated in (8). The example in (7) also shows that the scale of exclusion involves only relevant alternatives, i.e., the people ranked in terms of importance in (8), because (7) is true even if Maria talked to the manager’s secretary (assuming that this conversation was irrelevant to the topic of her conversation with the manager). Note that in English *merely* is dedicated to scalar readings, (6)–(7) (Coppock & Beaver 2011, 2014).

- (8) Scale for (6)–(7):
 - ‘Jan is the president.’
 - ‘Jan is a vice president.’
 - ...
 - ‘Jan is a senior manager.’
 - ‘Jan is a manager.’
 - ‘Jan is an assistant manager.’
 - ...
 - ‘Jan is an employee.’

The Polish exclusives in the example in (1), *tylko*, *zaledwie*, *dopiero* and *az*, are all compatible with the scalar readings, but now the examples in (9)–(12) will show

that Polish does not have one adverbial that can contribute the exclusive meaning in all contexts like English *only* can.² The quantificational reading requires *tylko*, as shown in the sentence in (9) which is the counterpart of the sentence we saw in (3). With temporal scales, on the other hand, *tylko* cannot be used, (10), unlike *only* in (5). With the scale of important people, (11)–(12), *tylko*, *zaledwie* and *dopiero* are possible, though here the manager is low on the scale, just like in (6)–(7) with *only/merely*.

- (9) Maria spotkała {tylko / #zaledwie / #dopiero / #aż} Jana.
 Maria met only merely only_{TEMP} Aż Jan
 ‘Maria met only Jan and nobody else.’
- (10) Jan przyjechał {dopiero / aż / #tylko / ?zaledwie} o 9-tej.
 Jan arrived only_{TEMP} Aż only merely at 9
 ‘Jan arrived only at 9 PM and no sooner.’
- (11) Jan jest {tylko / zaledwie / dopiero / #aż} menedżerem.
 Jan is only merely only_{TEMP} Aż manager
 ‘Jan is only/merely a manager (so far) and nobody more important.’
- (12) Maria rozmawiała {tylko / zaledwie / dopiero / #aż} z menedżerem.
 Maria talked only merely only_{TEMP} Aż with manager
 ‘Maria only/merely talked to the manager (so far) and nobody more important.’

Crucially, the scalar readings of (11)–(12) “flip” when *aż* is used: In (13) the manager is interpreted to be high on the scale of importance. The exclusive *aż* results in a reading that is similar to the effect of the addition of *even* (Polish *nawet*). In the sentence with *nawet* in (14), similarly to (13), the manager is implied to be a noteworthy person for Maria to talk to. However, *nawet* is not compatible with the continuation that Maria did not talk to anybody else, (14), in contrast to *aż* in (15). This is because *nawet/even* is an additive scalar particle (Horn 1969, Karttunen & Peters 1979, Rooth 1985, 1992, a.o.), whereas *aż* is an exclusive as I argued in Tomaszewicz (2012), Tomaszewicz (2013a,b).

- (13) Maria rozmawiała {#tylko / #zaledwie / #dopiero / aż} z menedżerem.
 Maria talked only merely only_{TEMP} Aż with manager
 ‘Maria talked to somebody so important as the manager.’

²All non-English examples in this chapter are from Polish, unless indicated otherwise.

- (14) Maria rozmawiała nawet z menedżerem, #ale nie rozmawiała z
 Maria talked even with manager but not talked with
 nikim innym.
 nobody else
 ‘Maria even talked to the manager, [*Incongruous*:] but she did not talk to
 anybody else.’
- (15) Maria rozmawiała aż z menedżerem, ale nie rozmawiała z
 Maria talked aż with manager but not talked with
 nikim innym.
 nobody else
 ‘Maria talked to somebody so important as the manager, but she did not
 talk to anybody else.’

I proposed in Tomaszewicz (2012), Tomaszewicz (2013a,b) that *aż* is the scalar opposite of *merely/zaledwie*: It asserts exclusion just like *only/merely*, evokes scales like *merely*, but it places the mentioned alternative high on the scale. In this chapter, I provide tests to specify the meaning components for each of the exclusives shown in (1), demonstrating that the semantics contains a common core predicting the existence of scalar opposites like *aż* cross-linguistically, even though such a lexical item is unavailable in, for instance, English.

Our starting point is the basic meaning contribution of *only/merely*, which involves, on the one hand, exclusivity (16a)–(16b) and, on the other hand, the preja-cent, which is the meaning of the sentence without *only/merely* (16) (Klinedinst 2005, Roberts 2006, van Rooij & Schulz 2007, Beaver & Clark 2008, a.o.).³

- (16) Maria only talked to the manager.

Exclusive/negative contribution:

- a. ‘Maria did not talk to anybody else than the manager.’

QUANTIFICATIONAL READING

- b. ‘Maria did not talk to anybody more noteworthy than a manager.’

SCALAR READING

³This use of the term PREJACENT was introduced by Horn (1996); however, Roberts (2006) observes that often with *only* the preja-cent does not correspond to any overt syntactic constituent. von Stechow & Heim (2002) use the term for the constituent that is the sister of the operator at the level of logical syntax. Roberts (2006) addresses the debate on the status of the preja-cent with *only* arguing that it is “closer to a presupposition than to an entailment or a conversational implicature”. Crnič (2013) argues that the mixed behavior (preja-cent presupposition vs. existential presupposition) results from two separate meaning components at LF with different scope.

Positive contribution:

c. ‘Maria talked to (at least) the manager.’

PREJACENT

The quantificational reading in (16a) can be seen as an instance of exclusion along a logical/entailment scale (Bonomi & Casalegno 1993, van Rooij 2002, Beaver 2004, Klinedinst 2005, Riester 2006, a.o.). The dimension of the scale is set relevant to the context, and where no such pragmatic ordering is readily available (such as the ranking of people in terms of importance), the logical scale is the default (e.g., all other people are excluded).

In the following sections I present a battery of tests to demonstrate that with all four Polish adverbial particles, *tylko*, *zaledwie*, *dopiero* and *aż*, exclusivity is asserted, the prejacent is presupposed, and scalarity, i.e., the ‘low/high on the scale’ reading, is presupposed. In addition to the exclusive assertion and the prejacent presupposition, it will be shown that scalarity comes from focus association, so that the focal presupposition supplies the set of alternatives for the ranking.

3 Focus association

Only belongs to so-called FOCUS-SENSITIVE adverbs because its surface position interacts with the placement of focus. The semantic effects of focus on the interpretation of *only* go hand in hand with syntactic/phonological effects, i.e., the way *only* modifies the meaning of the sentence depends on which constituent is focused, although the syntactic associate needs not always be marked by easily perceptible prosodic prominence. In Polish focus is marked prosodically, with pitch and intensity as the main acoustic correlates (Hamlaoui et al. 2019). The focused constituent is called the SYNTACTIC ASSOCIATOR. It can be a constituent that is immediately modified by *only*, as we have seen above and as shown below in (17): ‘only for a loan’.

- (17) Hanka poprosiła tylko [PP o [DP pożyczkę]], i o nic więcej /
 Hanka asked only for loan and for nothing more
 innego.
 else
 ‘Hanka only asked for a loan and nothing more/else.’

But the associator can also be more distant, as can be seen in (18). The different continuations in (18a)–(18d) result from different focus association configurations (associated with different prosodic patterns). Here *tylko* precedes the verb but it can associate with the object DP so that the sentence has the exact

same interpretation as in (17). It can also associate with the verb itself, as shown by the compatibility with the continuation in (18b), and it can associate with the whole VP, as evidenced by the continuation in (18c). In contrast to English, *tylko* in the preverbal position can also associate with the subject DP, (18d). When *tylko* precedes the subject it can, unsurprisingly, associate with it, (19a), but it can also associate with the whole sentence, (19b), which, again, is not possible in English.

- (18) [DP Hanka] tylko [VP [V poprosiła] o [DP pożyczkę]],
 Hanka only asked for loan
- a. ... i o nic innego nie poprosiła. OBJECT DP FOCUS
 and for nothing else not asked
 ‘Hanka asked only for a loan and did not ask for anything else.’
- b. ... a nie zażądała. V FOCUS
 and not demanded
 ‘Hanka only asked for a loan and did not demand it.’
- c. ... i nic innego nie zrobiła. VP FOCUS
 and nothing else not did
 ‘Hanka only asked for a loan and did not do anything else.’
- d. ... i nikt inny nie prosił. SUBJECT DP FOCUS
 and nobody else not asked
 ‘Only Hanka asked for a loan and nobody else did.’
- (19) (To) tylko [S [DP Hanka] poprosiła o pożyczkę],
 TOP only Hanka asked for loan
- a. ... i nikt inny nie prosił. SUBJECT DP FOCUS
 and nobody else not asked
 ‘Hanka only asked for a loan and nobody else did.’
- b. ... i nic innego się nie stało. SENTENCE FOCUS
 and nothing else self not happened
 ‘It is only that Hanka asked for a loan and nothing else happened.’

The associate determines the implicit comparison with alternatives of the same type. So when *tylko* in the sentence-initial position associates with the whole clause (sentence focus in (19b)), what is excluded are other things that could have happened that are ranked higher on the scale of importance. A typical context for (19b) is a question ‘Did anything happen?’ and an answer such as ‘No worries. It is just ...’.

In (20) I provide an example from a Polish comedy TV series “Alternatywy 4” (episode 9), from a scene where what appears to be a woman but is in fact a robot,

jumps down from a balcony, breaks her neck severing the head, then stands up and picks up the head. The people witnessing this are screaming in shock and a man is trying to calm them down saying that nothing out of the ordinary is happening.

- (20) To nic takiego, to nic takiego! To tylko ten psychiczny dostał
it nothing such it nothing such it only this lunatic got
szału!
amok
'It is nothing, it is nothing! It is just that lunatic going crazy!'

I am pointing out the differences between Polish and English because the syntax (and prosody) of focus association may vary considerably between languages. The extra options available in Polish are not limited to Slavic languages but are also possible in German (Büring & Hartmann 2001, Steindl 2013, 2017). In (21) German *nur* ‘only’ associates with the subject DP that precedes it, while in (22) *nur* is in the sentence-initial position and can associate either with the subject or the whole clause. In English, on the other hand, the associates of *only* must be in its surface c-command domain (Taglicht 1984, Beaver & Clark 2008, Toosarvandani 2010, a.o.).

- (21) [DP Juliane]_F hat nur die Blumen gegossen.
 Juliane has only the flowers watered
 ‘It was only Juliane that watered the flowers.’ (German; Steindl 2017)
- (22) Nur [s [DP Juliane]_{F_a} spielt Klavier]_{F_b}.
 only Juliane plays piano
 a. ‘Only Juliane is playing the piano.’ SUBJECT DP FOCUS
 b. ‘Nothing is going on except for Juliane playing the piano.’ SENTENCE FOCUS
 (German; Steindl 2013)

Of course, the different focus association patterns are associated with different prosodic contours of the sentence; specifically, the location of pitch accent and nuclear stress identify the constituents in focus. Without going into details of phonology, syntactic evidence for obligatory focus association comes from clitics. Clitics and reduced pronouns force a wider domain reading ((23b) vs. (23a)), which shows that *tylko* cannot associate with weak (unfocused) pronouns. In (23b) *tylko* has to associate with the adverbial or the verb, which can be independently focused (cf. Hoeksema & Zwarts 1991, Beaver & Clark 2008 on focus association of *only* in Dutch).

- (23) a. [_{Adv} Wieczorem] tylko widziałem [_{DP} niego].
 evening only saw him
 i. ‘In the evening I saw him and nobody else.’
 ii. ‘In the evening, and at no other time, I saw him.’
 b. [_{Adv} Wieczorem] tylko [_{VP} go [_V widziałem]].
 evening only him saw
 i. ‘In the evening I saw him and did not e.g. hear him.’
 ii. ‘In the evening, and at no other time, I saw him.’
 iii. *Unavailable*: ‘In the evening I saw him and nobody else.’

The syntactic tests above, the interaction between the size of the focused constituent and the syntactic position of *tylko* (18)–(19), and the inability of *tylko* to associate with non-focused elements such as clitics (23), suggest that *tylko* obligatorily associates with focus. The same can be shown for other Polish exclusives, as demonstrated for *aż* in Tomaszewicz (2012), Tomaszewicz (2013a,b).

4 Different ways to exclude alternatives

4.1 *tylko*

Only is standardly assumed to contribute both to the assertion and the presupposition of the sentence in which it occurs (Horn 1969, Klinedinst 2005, Roberts 2006, 2011, van Rooij & Schulz 2007, Beaver & Clark 2008). The asserted meaning can be either scalar (the dedicated use of *merely*) or quantificational. In (11)–(12) we saw that *tylko* like *only* can have both scalar and quantificational readings; (12) is repeated below in (24), and the examples for the two types of scales are given in (25). In (24) *tylko* associates with the focus on the PP ‘with the manager’.

- (24) Maria rozmawiała tylko z menedżerem.
 Maria talked only with manager
 ‘Maria only talked to the/a manager.’
 SCALAR READING: ‘Maria talked to nobody ranked higher than the/a manager.’
 QUANTIFICATIONAL READING: ‘Maria talked to nobody other than the/a manager.’

(25) Possible scales for (24):

a. Contextual

‘Maria talked with the president’
 ‘Maria talked with vice president’
 ...
 ‘Maria talked with a manager’
 ...
 ‘Maria talked with an employee’

b. Logical

‘Maria talked with 10 people’
 ‘Maria talked with 9 people’
 ...
 ‘Maria talked with 2 people’
 ‘Maria talked with 1 person’
 ‘Maria talked with nobody’

Quantificational *only* asserts that no other alternative is true, e.g., Maria only talked to the manager and nobody else. As noted in the previous section, the alternatives are either seen as not ordered or as an instance of exclusion along a logical/entailment scale, such as a numeral scale, as shown in (25b) (van Rooij 2002, Klinedinst 2005, Riester 2006, Coppock & Beaver 2011, a.o.). The dimension of the scale is set pragmatically, and in contexts where no pragmatic ordering is available, the logical scale is the default.

Using tests for presupposition, such as embedding under a question operator or negation (Chierchia & McConnell-Ginet 1990), we can identify three meaning components of *only*, one asserted (exclusivity) and two presupposed components (the prejacent and the ‘low on the scale’ reading).⁴ We will use these tests on the three meaning components of the sentence in (24) identified in (26a–c). The asserted component is the exclusion of all alternatives higher on the scale: the rank of the person that Maria talked to is no higher than a manager, (26a).⁵ Additionally, *only/merely* contributes two presuppositions, (26b), (26c). First, it presupposes that the alternatives under consideration are at least as high on the scale as the prejacent (Klinedinst 2005, Beaver & Clark 2008). Other relevant ranks of the person that Maria talked to are at least as high as the manager, (26b) (lower ranks are simply not relevant). The second presupposition is that the prejacent is low on the scale of alternative propositions, which I paraphrase as ‘is not significant’, (26c) (Klinedinst 2005).

⁴Note that there is extensive debate on the nature of the non-truth conditional, i.e., projective content. Many different subtypes have been discussed, including pragmatic and semantic presuppositions (Stalnaker 1973), conversational and conventional implicature (Grice 1981, Bach 1994, Potts 2005, 2015), F-implicature (Horn 2013), proffered content (Roberts 1996) and several other types and their groupings (e.g., Classes A–D in Tonhauser et al. 2013).

⁵The alternative ‘Maria talked to nobody’ is also excluded, which, as suggested by Radek Šimík, could be seen as lower, and not higher, on the scale. However, the logical/entailment scale on the quantificational reading is not one-dimensional but it is a mereological structure (Riester 2006, Coppock & Beaver 2011).

- (26) Maria rozmawiała tylko z menedżerem.
 Maria talked only with manager
 ‘Maria only talked to the manager.’
- a. Maria talked to nobody ranked higher than a manager. ASSERTION
 - b. Maria talked to at least a manager. PRESUPPOSITION
 - c. Maria’s talking to a manager is not significant. PRESUPPOSITION

Embedding under a question operator and under negation shows that the exclusive component in (26a) is an assertion: it is targeted both by the question operator, (27a), and by the negation operator, (28a). It is now readily apparent that the presupposition in (26b) projects in a question, (27b), and under negation, (28b). The low-on-the-scale inference in (26c) also clearly projects, (27c), (28c).

- (27) Czy Maria rozmawiała tylko z menedżerem?
 Q Maria talked only with manager
 ‘Did Maria only/merely talk to a manager?’
- a. Did Maria talk to anybody ranked higher than a manager? QUESTION
 - b. Maria talked to at least a manager. PRESUPPOSITION
 - c. Maria’s talking to a manager is not significant. PRESUPPOSITION
- (28) Maria nie rozmawiała tylko z menedżerem.
 Maria not talked only with manager
 ‘Maria did not merely talk to a manager.’
- a. Maria talked to somebody ranked higher than a manager.
 NEGATIVE ASSERTION
 - b. Maria talked to at least a manager. PRESUPPOSITION
 - c. Maria’s talking to a manager is not significant. PRESUPPOSITION

Beaver & Clark (2008) and Zeevat (2009) call the low-on-the-scale reading of *only* MIRATIVE. As a mirative marker *only* implies that the prejacent falls short of what is expected. This reading is easily detected with numerals, for instance, if you hear *The driver is only 1 stop away*, you know that 1 stop is very close. However, the sentence *The driver is only 15 stops away* also tells you that the driver is close – contrary to what you would expect given the usual duration of 15 stops (so for this statement to be true these 15 stops must be very short). Without *only*, the sentence *The driver is 15 stops away* does not create any expectation whether the driver is going to reach you soon or not.

Consider also that *tylko* can scope both below and above negation, as shown in (29). Both readings can be detected: quantificational, (29a), (29b), and scalar, (29c), (29d).

- (29) Maria nie rozmawiała tylko z menedżerem.
 Maria not talked only with manager
- a. 'Maria talked to somebody else in addition to the manager.' NOT > ONLY
 - b. 'Maria talked to everyone else except the manager.' ONLY > NOT
 - c. 'Maria talked to somebody more noteworthy than the manager in addition to the manager.' NOT > ONLY
 - d. 'Maria talked to everyone more noteworthy than the manager except the manager.' ONLY > NOT

Another test to distinguish between asserted and presupposed content are EMOTIVE FACTIVE predicates, such as 'disappointed' in (30) (Chierchia & McConnell-Ginet 1990, Beaver & Clark 2008, Simons et al. 2011, Coppock & Beaver 2014). These predicates target the asserted negative component with *tylko*, showing, again, that exclusivity is asserted and scalarity presupposed, (30).

- (30) Jestem zawiedziony, że Maria rozmawiała tylko z menedżerem.
 AUX disappointed that Maria talked only with manager
 'I am disappointed that Maria did not talk to anybody else/more noteworthy.'

Example (30) nicely illustrates the contribution of the preajacent presupposition 'Maria talked to at least a manager', because clearly the speaker is not disappointed that Maria did talk to the manager. A further test, utterance of assent/dissent (Roberts 2011) is shown in (31): The exclusive component is asserted (targeted by direct affirmation) and the preajacent projects.

- (31) A: Maria rozmawiała tylko z menedżerem.
 Maria talked only with manager
 'Maria only talked to the manager.'
- B: To prawda. {Nie rozmawiała z nikim innym. / #Rozmawiała
 this true not talked with nobody else talked
 z menedżerem.}
 with manager
 'That is true. {She did not talk to anybody else. / [Incongruous:] She talked to the manager.'}

- B': To prawda. {Nie rozmawiała z kimś ważniejszym. /
 this true not talked with somebody more.important
 #Rozmawiała z menedżerem.
 talked with manager
 'That is true. {She did not talk to somebody more noteworthy. /
 [Incongruous:] She talked to the manager.'}

In the diagnostics above, *tylko* patterns with *only*, in that it has both scalar and quantificational readings and its exclusive component is asserted, while the pre-jacent and the low-on-the-scale components are presupposed. In fact, *tylko* appears to be the most likely candidate for the counterpart of *only*, since it is the only exclusive that can have quantificational readings (in contrast to *zaledwie*, *dopiero* and *aż* discussed in the following sections). However, there is a significant difference between *tylko* and *only*: Although *tylko* has scalar readings, it is incompatible with a temporal scale. The sentence in (32a) can only mean that Marek woke up once (excluding all other 'times of waking up', i.e., the quantificational reading). The 'no later than'-reading is available with (*za*)*ledwie*, *dopiero* and *aż* but not with *tylko* (32b).

- (32) a. Martwi mnie, że Marek obudził się tylko o godzinie 13.
 worries me that Marek woke self only at hour 13
 'I am worried that Marek woke up just once at 1 PM.'
- b. Martwi mnie, że Marek obudził się {zaledwie / dopiero / aż /
 worries me that Marek woke self merely only_{TEMP} AŻ
 #tylko} o godzinie 13.
 only at hour 13
 'I am worried that Marek woke up only at 1 PM and no sooner.'

The contrast between (33a) and (33b) shows that the presence of *tylko* results in an anomaly, which indicates how strong the restriction on the type of scale is – it is not possible to coerce a temporal reading.

- (33) a. Jest {dopiero / zaledwie} luty.
 is only_{TEMP} merely February
 'It is only February.' ('It is no later than February.')
- b. # Jest tylko luty.
 is only February
 Incongruous: 'It is no other month now than February.'

If the dimension of the scale is underspecified in the semantics of an exclusive adverb, we expect it to be easily manipulated by the context (the semantics of the focused constituent the adverb associates with, e.g. ‘February’ above, and the pragmatics). The four Polish exclusives, however, appear to be more contextually restricted. Distance, temporal and numeral scales are all natural, inherent, dimensional scales, so the prediction should be that an exclusive that works with any kind of a scale is available in any language. This seems to be the case for *only* in English:

- (34) We only reached Wilson Peak on that trip and were turned back [...] by an ice storm.
- (35) I am usually up at 8:30am, but this morning for some reason I only woke up at 10am.
- (36) Brussels was ranked only 62nd in Mercer’s list of the world’s most expensive cities for expats.

If Polish *tylko* can act scalar with individuals ordered on a contextual scale (importance, etc.) why is it incompatible with a temporal scale? In the next sections we will see that *zaledwie* is dedicated to pragmatic scales (just like *merely*) and *dopiero* to temporal scales. This means that *dopiero* might be blocking *tylko* (but not *zaledwie*) with a temporal scale, and that *zaledwie* does not block *tylko* with contextual scales (which perhaps suggests an interplay between the conventional nature of a scale and frequency in language use given that *zaledwie* is less frequent than *tylko*).

4.2 (za)ledwie

While we saw in the previous section that *tylko* has both quantificational and scalar readings, *zaledwie* is specified for scalar readings – it is a counterpart of English *merely*. What makes it particularly interesting though, is that it is the exact scalar opposite of the exclusive *aż*, which we will see in the next section. The example in (37) shows that *zaledwie* disprefers quantificational readings: In an impoverished context where no pragmatic ranking for Maria and others is available, the default is a scale of numbers of people who came (recall the two options for two scales in (25) above) and so the use of *tylko* is preferred.

- (37) {Tylko / ?Zaledwie} Maria przyszła.
only merely Maria came
‘Only/Merely Maria came.’

- a. QUANTIFICATIONAL READING: ‘Nobody other than Maria came.’
- b. UNAVAILABLE SCALAR READING: ‘Nobody ranked higher than Maria came.’

Accordingly, (38) shows that the most accessible interpretation for *zaledwie* is that the excluded alternatives involve ranks higher than the manager (38B') and not people other than the manager (38B).

- (38) A: Maria rozmawiała zaledwie z menedżerem.
 Maria talked merely with manager
 ‘Maria only talked to the manager.’
- B: To prawda. {#Nie rozmawiała z innymi. / #Rozmawiała z
 this true not talked with others talked with
 menedżerem.}
 manager
 ‘That is true. {[Incongruous:] She did not talk to other people. /
 [Incongruous:] She talked to the manager.’
- B': To prawda. {Nie rozmawiała z kimś ważniejszym. /
 this true not talked with somebody more.important
 #Rozmawiała z menedżerem.}
 talked with manager
 ‘That is true. {She did not talk to somebody more noteworthy. /
 [Incongruous:] She talked to the manager.’

Like *tylko*, *zaledwie* can scope both below and above negation, (39), which is expected given that *tylko* can do so on both scalar and quantificational readings.

- (39) Maria nie rozmawiała zaledwie z menedżerem.
 Maria not talked merely with manager
- a. ‘Maria talked to somebody more noteworthy than the manager in addition to the manager.’ NOT > ONLY
 - b. ‘Maria talked to everyone more noteworthy than the manager except the manager.’ ONLY > NOT

The emotive factive diagnostic, just like with *tylko* in (28), reveals the positive presupposition of the prejacent, (40) (i.e., the speaker is not disappointed that Maria talked to the manager.)

- (40) Jestem zawiedziony, że Maria rozmawiała zaledwie z menedżerem.
AUX disappointed that Maria talked merely with manager
'I am disappointed that Maria did not talk to anybody more noteworthy
than the manager.'

What is interesting about *zaledwie* is that of the four exclusive particles, it is the only one that can work as the counterpart of English *barely* and receive APPROXIMATIVE (41) (Roberts 2011) and COUNTERFACTUAL (42) (Toosarvandani 2010) readings. On approximative readings, (41), the prefix *za-* can be dropped. On counterfactual readings, (42), *ledwie* occurs without *za-*.

- (41) {(Za)ledwie / #Tylko / #Dopiero / #Aż} Marek otworzył drzwi, a
merely only only_{TEMP} AŻ Marek opened door and
zadzwoił telefon.
rang telephone
'Marek barely opened the door, when the telephone rang.'
- (42) Marek {(*)za)ledwie / #tylko / #dopiero / #aż} otworzył drzwi.
Marek merely only only_{TEMP} AŻ opened door
'Marek barely opened the door. (It was more likely for him to not open
the door.)'

Ledwie is syntactically an adverbial (also *ledwo*), i.e. can only modify the VP, as shown by the contrast in (43): In (43a) *(za)ledwie* associates with Berlin, in (43b) *ledwie* associates with the VP 'reached Berlin' as can be seen from the translation, and here the prefix *za-* is not possible.

- (43) a. Dotarłem (za)ledwie do Berlina.
reached merely to Berlin
'I only reached Berlin.'
- b. (*Za)ledwie dotarłem do Berlina.
merely reached to Berlin
'I barely made it to Berlin.'

Summing up, the exclusive *zaledwie* prefers scalar readings (like *merely* does), but its variant *ledwie* has approximative and counterfactual readings.

4.3 *dopiero*

In (32a) above we saw that 'It is only February' requires *dopiero* in its translation, repeated below in (44), because *tylko* results in an anomaly ('It is no other month

now than February'). This suggests that *dopiero* is a specialized TEMPORAL *only* and I gloss it as *only*_{TEMP}.

- (44) a. It is only February.
 b. Jest {*dopiero* / **tylko*} luty.
 is *only*_{TEMP} only February
 'It is only February. (It is no later than February.)'

The use of *dopiero*, in contrast to the other exclusives, coerces the scale associated with the predicate, e.g. quantity in (45)–(46), to an interpretation with respect to time.

- (45) Maria rozmawiała *dopiero* z 2 osobami.
 Maria talked *only*_{TEMP} with 2 people
 'Maria only talked to 2 people so far.'

While in the example (46) below temporal alternatives are obligatorily evoked, which results in the reading 'so far Jan bought 2 tomatoes and he will buy more', there is no such expectation with *tylko* in (47).

- (46) Jan kupił *dopiero* 2 pomidory.
 Jan bought *only*_{TEMP} 2 tomatoes
 'Jan bought only two tomatoes and we expect him to buy more.'
- (47) Jan kupił *tylko* 2 pomidory.
 Jan bought *only* 2 tomatoes
 'Jan bought only two tomatoes.'

Logically, the temporal alternatives should be later in time: In (45) more people will come later, in (46) Jan will buy more tomatoes later, so in a context where such alternatives are not possible, as in (48) where Jan is not going to write any more, *dopiero* is infelicitous.

- (48) Jan skończył pisać swój esej. Ma {*tylko* / #*dopiero*} 10 stron.
 Jan finished writing his essay has *only* *only*_{TEMP} 10 pages
 'Jan finished writing his essay. It is only 10 pages long.'

The example above provides an interesting contrast to the sentence in (1) where the team ended up in 30th place (i.e., was not going to compete anymore) and *dopiero* is felicitous. Indeed, we find *dopiero* with scales that are not (necessarily) correlated with temporal progression, such as the example in (49) about a partial achievement, idiomatically referred to as halfway to success.

- (49) Dobra drukarka to jednak dopiero połowa sukcesu. Drugą, równie
 good printer is however only_{TEMP} half success second equally
 istotną częścią jest odpowiedni toner.
 important part is suitable toner
 'A good printer is only half of the success. Another equally important
 thing is a good toner.'

Like the other exclusives we considered, *dopiero* can take scope both above and below negation as we can see in (50)–(51).⁶

- (50) Dopiero 2 osoby przyszły.
 Only_{TEMP} 2 people came
 'So far only 2 people showed up.'
- (51) Nie przyszły dopiero 2 osoby.
 not came only_{TEMP} 2 people
 a. 'More than 2 people came.' NOT > ONLY
 b. 'Everyone came except 2 people.' ONLY > NOT

The translations for the examples above indicate that *dopiero* is similar to the temporal adverbs *so far* (*jak dotąd*) / *yet* (*jeszcze*). However, *dopiero* also comes with an exclusive component which these temporal adverbs do not have by themselves. First, let us note that *dopiero*, like *yet*, and unlike *so far*, can occur with negation and in questions, (52). Note also that *so far* is not an NPI in contrast to *only* and *yet*, (53).

⁶Native speakers of Polish can reflect on how additional context facilitates the interpretation of the two scope configurations. Consider also the different prosodic contours with different focus structures. In (51a) *dopiero* associates with focus on '2 people', see (i.a) below. In (51b) *dopiero* takes wide scope and associates with sentential focus, which can be made overt with scrambling as in (i.b).

- (i) a. Nie, nie przyszły dopiero [2 osoby]_F, są już [3 osoby]_F.
 no not came only_{TEMP} 2 people are already 3 people
 'So far not just 2 people showed up, but 3.' NOT > ONLY
- b. Wydaje mi się, że {nie przyszły dopiero 2 osoby / dopiero [2 osoby nie
 seems me self that not came only_{TEMP} 2 people only_{TEMP} 2 people not
 przyszły]_F}, ale może się pomyliłam w liczeniu.
 came but maybe self mistook in counting
 'I think that only 2 people didn't show up, but maybe I counted wrong' ONLY > NOT

- (52) Kto wie, czy to nie jest dopiero początek problemów
 who knows if this not is only_{TEMP} beginning problems
 premiera?
 prime.minister.GEN
 ‘Who knows if this is only the beginning of the prime minister’s
 problems?’
- (53) Who knows, maybe this is {only/yet/*so far} the beginning of the prime
 minister’s problems?

Secondly, with *so far* in (54a) there is an implication that Maria talked to just one person, but this is not an assertion because it is possible to continue the sentence as in (54b) in contrast to (54c) containing an exclusive. The counterpart of (54c) in Polish, (55), is compatible with both *tylko* and *dopiero*.

- (54) a. So far Maria has talked to the manager.
 b. So far Maria has talked to the manager and a couple of other people.
 c. So far Maria has only talked to the manager #and a couple of other people.
- (55) Jak dotąd Maria rozmawiała {dopiero / tylko} z menedżerem.
 how till.now Maria talked only_{TEMP} only with manager
 ‘So far Maria has only talked to the manager.’

That *so far* does not assert the exclusion of alternatives is also apparent in (56)–(58) below. In (56) we can expect that the person will call later, and accordingly in the Polish translation in (57) *dopiero* is not felicitous.

- (56) So far he has not called.
- (57) {Jak dotąd / *dopiero} nie zadzwonił.
 how till.now only_{TEMP} not called
 ‘So far he has not called.’ (Lit. ‘Up till now he has not called.’)

The presence of *so far* and *only* in (58) together results in an implication that documenting 373 species is a lot, because ‘it is still only February’, as said in 2012, indicates that this number was reached earlier than expected.⁷

⁷The source of (58) is <https://james-champion.com/friday-24th-february-2012/> (accessed on 29 March 2012). In fact, John’s initial goal for 2012 was 500 species and by the end of the year he documented 591 species.

- (58) John Cahill is going for an all-out mega-bird list for Guatemala in 2012.
[...]. So far he has reached 373 species in Guatemala this year – and it is
still only February!

In contrast to *so far* in (58), *dopiero* in the sentence in (59) contributes the reading that 373 species is not much, because, as the utterance of assent test shows, (60), the exclusive component ‘no more than 373 yet’ is asserted. The same test shows that *jak dotąd* (*so far*) lacks the exclusive component.

- (59) A: Jan dotarł dopiero do 373 gatunków.
Jan reached only_{TEMP} to 373 species
‘Jan has only reached 373 species so far.’
B: To prawda. {Nie dotarł do więcej niż 373 gatunków. / #Dotarł
this true not reached to more than 373 species reached
do 373 gatunków.}
to 373 species
‘That’s true. {He hasn’t reached more than 373 species. /
[Incongruous:] He has reached 373 species.’
- (60) A: Jan dotarł jak dotąd do 373 gatunków.
Jan reached so far to 373 species.
‘Jan has so far reached 373 species.’
B: To prawda. Dotarł (już) do 373 gatunków.
this true reached already to 373 species
‘That’s true. He has (already) reached 373 species.’

The data above demonstrates that *dopiero* specializes for temporal scales and now we will see that the diagnostics for assertion and presupposition replicate the results we found for the other exclusives. The utterance of assent test, (61), the emotive factive predicate test, (62) and negation test, (63), show that the exclusion of alternatives higher on the (temporal) scale is asserted, the prejacent is presupposed and the low-on-scale reading is presupposed.

- (61) A: Maria rozmawiała dopiero z menedżerem.
Maria talked only_{TEMP} with manager
‘Maria only talked to the manager so far.’
B: To prawda. {Nie rozmawiała (jeszcze) z innymi. / #Rozmawiała
this true not talked yet with others talked
z menedżerem.}
with manager
‘That is true. {She has not talked to other people yet. / [Incongruous:]
She talked to the manager.’

B': To prawda. {Nie rozmawiała (jeszcze) z nikim ważniejszym. /
 this true not she.talked yet with nobody more.important
 #Rozmawiała z menedżerem.
 she.talked with manager
 'That is true. {She has not talked to anybody more noteworthy yet. /
 [Incongruous:] She talked to the manager.'}

- (62) Jestem zawiedziony, że Maria rozmawiała dopiero z menedżerem.
 AUX disappointed that Maria talked only_{TEMP} with manager
 'I am disappointed that Maria hasn't talked to anybody else/more
 noteworthy yet.' (Implied: 'Maria talked to the manager.')
- (63) Maria nie rozmawiała dopiero z menedżerem.
 Maria not talked only_{TEMP} with manager
- 'Maria has already talked to somebody more noteworthy than the
 manager in addition to the manager.' NOT > ONLY
 - 'Maria has already talked to everyone more noteworthy than the
 manager except the manager.' ONLY > NOT

Finally, we must note that while *dopiero* excludes stronger alternatives on a temporal scale, they are not necessarily later, as is clear in (64). What is excluded in (64a) are earlier alternatives for finishing the writing. In this context, the exclusive that I am going to present in the next section, *aż*, is also possible. The fact that in (64b) *aż* is not felicitous, but instead it requires the preposition *do*, lit. 'to', (64c), which carries the meaning of 'until' (i.e., here *aż* is optional but *do* is not; see also (104) in Section 6) indicates that *aż* is not inherently specified for temporal scales and what allows it in (64a) is the presence of the numeral.

- (64) a. Skończę pisać {dopiero / aż / #tylko / #zaledwie} za 2 miesiące.
 will.finish writing only_{TEMP} AŻ only merely in 2 months
 'I will finish writing no sooner than in 2 months.'
- b. Skończę pisać {dopiero / #aż / #tylko / #zaledwie} w sobotę.
 will.finish writing only_{TEMP} AŻ only merely on Saturday
 'I will finish writing no sooner than on Saturday.'
- c. Skończę pisać {aż / #dopiero / #tylko / #zaledwie} do soboty.
 will.finish writing AŻ only_{TEMP} only merely until Saturday
 'I will finish writing until Saturday, which is late.'

We can conclude that *dopiero* has the same meaning components as *tylko/zaledwie*, but it is specified for numeral, especially temporal scales. In the next section, I will show that *aż* is a scalar opposite of *dopiero*.

4.4 *aż*

The exclusive *aż* is a scalar opposite of *zaledwie* (Tomaszewicz 2012, Tomaszewicz 2013a,b). The assent test in (65) shows us first that *aż* has no quantificational reading. Neither of the continuations in (65B) are possible – just like with *zaledwie* in (38B). Secondly, we see in (65B') that the asserted component is 'She did not talk to somebody less noteworthy', which is the direct opposite of what we saw in (38B') with *zaledwie*: 'She did not talk to somebody more noteworthy'.

- (65) A: Maria rozmawiała *aż* z menedżerem.
 Maria talked *AŻ* with manager
 'Maria talked to somebody so important as the manager.'
- B: To prawda. {#Nie rozmawiała z nikim innym. / #Rozmawiała
 this true not talked with nobody other talked
 z menedżerem}.
 with manager
 'That is true. {[Incongruous:] She did not talk to other people. /
 [Incongruous:] She talked to the manager.'}
- B': To prawda. {Nie rozmawiała z kimś mniej ważnym. /
 this true not talked with somebody less important
 #Rozmawiała z menedżerem}.
 talked with manager
 'That is true. {She did not talk to somebody less noteworthy. /
 [Incongruous:] She talked to the manager.'}

Native speaker intuitions are particularly robust for the juxtaposition of *aż* and *zaledwie* in (66)–(69) where the reason clause identifies the asserted content (Dretske 1972, Beaver & Clark 2008). In (66)–(67) the reason for Maria's (un)happiness is the position of the manager on the scale: high with *aż*, low with *zaledwie*. In (68)–(69) the reason for the party's success/failure is the fact that the expected number of guests was either exceeded, (68), or not achieved, (69). With *aż* in (66) Maria is happy because she is not talking to somebody less important than the manager and in (68) the party was a success because no less than 15 people came. The mere fact that Maria is talking to the manager is not the reason for her happiness – the fact that the manager is high on the scale of the relevant people is the reason. Similarly, the fact that 15 people came to the party is not the reason for the success – the fact that this number exceeded the expectations is the reason. Conversely, with *zaledwie* in (67) Maria is unhappy because her goal is to talk to someone more important than the manager and in (69) the party was a failure

because more than 15 people were expected. The reason clause test demonstrates that exclusivity is asserted, which means that *aż* works like the scalar opposite of *zaledwie*, *tylko* and *dopiero*: It carries a high-on-the-scale presupposition.

- (66) Maria jest zadowolona, bo rozmawia, aż z menedżerem.
 Maria is happy because talks AŻ with manager
 ‘Maria is happy because she is not talking to somebody less important than the manager.’
- (67) Maria jest niezadowolona, bo rozmawia zaledwie z menedżerem.
 Maria is unhappy because talks merely with manager
 ‘Maria is unhappy because she is not talking to somebody more important than the manager.’
- (68) Impreza się udała, bo przyszło aż 15 osób.
 party self succeeded because came AŻ 15 people
 ‘The party was a success because no less than 15 people came.’
- (69) Impreza się nie udała, bo przyszło {tylko / zaledwie} 15 osób.
 party self not succeeded because came only merely 15 people
 ‘The party was not a success because no more than 15 people came.’

A parallel example with *dopiero* does not work, because it evokes a temporal alternative when more people come.

- (70) Impreza się (nie) udała, #bo przyszło dopiero 15 osób.
 party self not succeeded because came only_{TEMP} 15 people
 ‘The party was (not) a success [*Incongruous:*] because so far only 15 people came.’

The high-on-the-scale presupposition found with *aż* projects under negation and in questions, and the orientation of the scale stays the same, as shown in (71c), (72c), (73c). Embedding under question and negation operators clearly reveals the asserted exclusive component – the scalar opposite of the contribution of *only*. Although (71) could simply be taken to convey that Maria talked to a manager and that this is significant, its negative version, (72), crucially asserts not only that Maria did not talk to a manager, but also that she talked to somebody lower on the scale than the manager, (72a). This is evidence for an asserted exclusive component.

- (71) Maria rozmawiała aż z menedżerem.
 Maria talked AŻ with manager
 ‘Maria talked to somebody so important as the manager.’
- a. Maria talked to nobody ranked lower than a manager. ASSERTION
 - b. Maria talked to at most a manager. PRESUPPOSITION
 - c. Maria’s talking to a manager is significant. PRESUPPOSITION
- (72) Maria nie rozmawiała aż z menedżerem.
 Maria not talked AŻ with manager
 ‘Maria talked to somebody less important than the manager.’
- a. Maria talked to somebody ranked lower than a manager. ASSERTION
 - b. Maria talked to at most a manager. PRESUPPOSITION
 - c. Maria’s talking to a manager is significant. PRESUPPOSITION

Exclusivity also emerges in the question in (73) which asks whether the position of the person Maria talked to is lower than a manager, (73a). (This is exactly the opposite of the effect of the presence of scalar *tylko* ‘only’ in the question in (28)).

- (73) Czy Maria rozmawiała aż z menedżerem?
 Q Maria talked AŻ with manager
 ‘Did Maria talk to somebody so important as the manager?’
- a. Did Maria talk to somebody ranked higher than a manager?
 QUESTION
 - b. Maria talked to at most a manager. PRESUPPOSITION
 - c. Maria’s talking to a manager is significant. PRESUPPOSITION

The exclusive component in (71a), (72a) and (73a), i.e. the assertion that the alternatives lower on the scale are excluded, is the reverse of the contribution of *only/merely*, where higher alternatives are excluded, as we saw in (26a), (27a) and (28a). The two other meaning components of *aż* that project are scalar reversals of the two presuppositions of *only/merely* identified in (26b/c), (27b/c) and (28b/c).

In (71b), (72b), (73b), the presupposition is that the alternatives under consideration can be at most as high on the scale as the prejacent. The presupposition in (71c), (72c), (73c) is that the prejacent is significantly high on the scale. For (71) this gives us the interpretation that considering the alternative positions that the person Maria talked to could hold (71b), being a manager is the most significant

among them (71b), (71c), and that that person is indeed no less than a manager (71a).

Beaver & Clark (2008) sum up the contribution of *only* as “contra expectation, nothing stronger holds” (p. 279). The scalar reversal of each of the meaning components of *only*, yields the interpretation for *aż* that can be described as ‘contra expectation, nothing weaker holds’.

It is frequently noticed that *even* functions as an antonym of *only* at the level of presupposition (Horn 1969, König 1991, Beaver & Clark 2008, Greenberg 2022). The following contrasts between *aż*, *zaledwie* and *nawet* ‘even’ show that the scalar presupposition of *aż* and *zaledwie* picks up the contextually salient scale, namely, distance, while *even* requires the likelihood scale (contra the speculation about *aż* vs. *nawet* in Bajaj 2016). Negation in the presence of *aż* in (74) conveys that the distance that Jan traveled was shorter than to Warsaw; the sentence cannot mean that Jan did not travel at all, which is a possible interpretation in (12) containing negation and *even*. The sentence in (74) tells us both that (i) Jan went as far as Łódź, but not Warsaw (see Figure 1 below for the illustration of the distance), and that (ii) Jan’s trip was not noteworthy, because it was not far. This meaning results from negation scoping above *aż* and is exactly parallel to the meaning of (76) where negation scopes above *zaledwie*. (76) says that (i) Jan went further than Łódź, Jan went as far as Warsaw, and that (ii) Jan’s trip was noteworthy, because it was far. Both the use of *aż* and *zaledwie* induce a distance reading in contrast to *nawet* ‘even’ in (75). In (75) *nawet* needs to scope above negation, the scalar presupposition is that the prejacent ‘Jan went to Łódź’ is the most likely among the alternatives (i.e., the least significant).

- (74) Jan nie pojechał aż do Warszawy, {pojechał zaledwie do Łodzi / #został
Jan not drove *Aż* to Warsaw drove merely to Łódź stayed
w domu}.
at home
‘Jan did not drive as far as Warsaw, {he drove to Łódź / [*Incongruous*:] he
stayed at home}.’
- (75) Jan nie pojechał nawet do Łodzi, {#pojechał zaledwie do Warszawy. /
Jan not drove even to Łódź drove merely to Warsaw
został w domu}.
stayed at home
‘Jan did not even drive to Łódź, {[*Incongruous*:] he drove merely to
Warsaw / he stayed at home}.’

- (76) Jan nie pojechał zaledwie do Łodzi, {pojechał aż do Warszawy. / #został Jan not drove merely to Łódź drove AŻ to Warsaw stayed w domu}.
at home
'Jan did not go merely to Łódź, {he went all the way to Warsaw / [Incongruous:] he stayed at home}.'

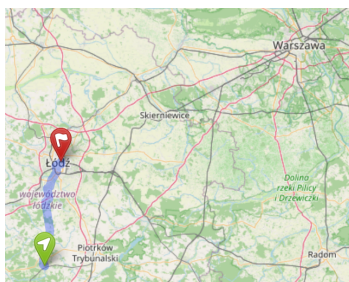


Figure 1: An illustration of the distances in the situation in example (74): 'Jan did not go as far as Warsaw, he went merely to Łódź.'

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In addition to its use as a scalar particle, *aż* functions in Polish as a complementizer, equivalent to *until* in English, (77a). As a complementizer, it is obligatorily sentence-initial, but as we saw in Section 3 focus associators can associate with foci of all sizes, including the whole sentence in Polish. I propose that the lexical semantics of *aż* is the same when it is interpreted as *until*, (77a)–(77b), and when it is a scalar adverb, (77c), but the syntax determines whether the ordering of alternatives is on a temporal scale or a pragmatic scale.⁸ I observe that when the main clause explicitly mentions duration, *aż* can either work as *until*, (77b), or as the scalar exclusive in which case it is preceded by the complementizer *that*, (77c). In (77c) *aż* can either associate with the subject so that the alternatives are people who could tell Ann to stop singing, or with the whole clause so that alternatives are other less serious things that may happen, e.g., the whole auditioning committee laughing.

- (77) a. Anna śpiewała, aż [_{IP} [_{DP} Maria] kazała jej przestać].
Anna sang AŻ Maria ordered her stop
'Anna sang until Maria told her to stop.'

⁸I thank Radek Šimík for pointing out that the other exclusive particles cannot switch in a parallel way with respect to the syntax, suggesting that the semantics of *aż* alone cannot be responsible for its function as a complementizer. This is an interesting topic for (a cross-linguistic) investigation.

- b. Anna śpiewała tak długo, aż [IP[DP Maria] kazała jej przestać].
 Anna sang so long AŻ Maria ordered her stop
 ‘Anna kept on singing until Maria told her to stop.’
- c. Anna śpiewała tak {słabo / długo}, że aż [IP[DP Maria] kazała jej przestać].
 Anna sang so badly long that AŻ Maria ordered her stop
 stop
- i. ‘Anna sang so badly/long, that out of all things that could happen Maria told her to stop.’
 - ii. ‘Anna sang so badly/long, that out of all people Maria told her to stop.’

In this section we established that *aż* is a scalar opposite of *zaledwie*, scalar *tylko* and *dopiero*. Similarly to *dopiero*, it receives a temporal interpretation when in the complementizer position. In the next section, I show how the lexical entries for the four exclusives can be related by a common core semantics.

5 Analysis

I propose that the four exclusives, *tylko*, *zaledwie*, *dopiero* and *aż* are a family related by having the same three semantic components: the exclusive assertion, the scalar presupposition and the prejacent/factive presupposition. The differences between them involve the direction of the scale and extra requirements on the sets of alternatives resulting in their idiosyncrasies. The facts are summarized in Table 1. Below I define the lexical entries for each member of the family in such a way that the common core of the three components is present and only the necessary parameters are tweaked.

The meaning of the scalar exclusive, *tylko/zaledwie* involves the assertion that no higher alternative is true, and the factive presupposition (78a), as well as the low-on-the-scale presupposition (78b). Scalarity is thus both asserted and presupposed.

(78) $\llbracket \text{tylko/zaledwie} \rrbracket^w(C)(p)$ is defined iff

- a. $\exists q \in C[q \geq p \wedge q(w) = 1]$
 [Factivity: an alternative at least as strong as *p* is true]
 and
- b. $\forall q \in C[q \neq p \rightarrow q > p]$
 [Scalarity: *p* is the weakest alternative]

Table 1: The meaning components of *tylko*, *zaledwie*, *dopiero* and *aż*

<i>tylko</i>	<i>(za)ledwie</i>	<i>dopiero</i>	<i>aż</i>
STATUS OF PREJACENT <i>p</i>			
<i>p</i> presupposed	<i>p</i> presupposed	<i>p</i> presupposed	<i>p</i> presupposed
SCALAR PRESUPPOSITION (<i>p</i> HIGH/LOW ON SOME SCALE <i>S</i>)			
<i>p</i> low on <i>S</i>	<i>p</i> low on <i>S</i>	<i>p</i> low on <i>S</i>	<i>p</i> high on <i>S</i>
EXCLUSIVE ASSERTION (ALTERNATIVES EXCLUDED, HIGHER/LOWER ON <i>S</i> THAN <i>p</i>)			
higher than <i>p</i>	higher than <i>p</i>	higher than <i>p</i>	lower than <i>p</i>
IDIOSYNCRASIES			
<i>S</i> is logical or pragmatic but not temporal	<i>S</i> is pragmatic; approximative readings	<i>S</i> is numeral/temporal	<i>S</i> is pragmatic; temporal ‘until’ meaning

if defined then

$$\llbracket \textit{tylko/zaledwie} \rrbracket^w(C)(p) = \forall q[q \in C \wedge q \neq p \wedge q > p] \rightarrow q(w) = 0]$$

The FACTIVE presupposition is the positive meaning component that projects under negation, therefore it is not just the prejacent proposition, but an alternative proposition that is at least as strong on the relevant scale (I am following Klinedinst 2005 and Coppock & Beaver 2014, but see also Beaver & Clark 2008 and Zeevat 2009 for a mirative presupposition that the prejacent falls short of what is expected). This positive contribution is paraphrasable with ‘at least’ as we first saw in the example in (16): ‘Maria only talked to the manager’ implies that ‘Maria talked to (at least) the manager’, (16c). Coppock & Beaver (2014) point out that this component is easier to see on scalar readings, in examples such as (79), than on quantificational readings, as in (80).

- (79) This is not {just/a mere} down payment.

Positive contribution: ‘This is at least a down payment.’

SCALAR READING Coppock & Beaver 2014: (8–9) p. 374

- (80) Mary did not invite only John and Mike.

Positive contribution: ‘Mary invited at least John and Mike.’

QUANTIFICATIONAL READING Coppock & Beaver 2014: (18) p. 379

Since my analysis follows the accounts where the quantificational and scalar readings are derived from the same lexical entry (Bonomi & Casalegno 1993, van

Rooij 2002, Beaver 2004, Klinedinst 2005, Riester 2006, Beaver & Clark 2008, Coppock & Beaver 2013, 2014), the factive presupposition of *tylko/zaledwie* in (78a) allows for both readings. As mentioned in Section 2, the dimension of the scale is set by the context and quantificational readings arise with logical scales, whereas scalar readings result from pragmatic orderings of alternatives in *C*. In impoverished contexts the logical scale is the default.

Greenberg (2022), building on Guerzoni (2003), proposes that the scalar presupposition of *only* is “superlative” so that *p* is the weakest alternative in the context of focus alternatives, *C*. She also discusses how a general notion of stronger alternatives, symbolized as ‘>’, suffices, given that *C* is similar to domain restriction on quantifiers and thus contains only contextually relevant alternatives, but at the same time the focus structure is constrained by the Question Under Discussion that is salient in the context (Rooth 1992, Roberts 1996, Beaver & Clark 2008, Coppock & Beaver 2013, 2014). Following Guerzoni (2003), Greenberg (2022) treats *only* and *even* as antonyms, with mirroring superlative presuppositions, but adds special constraints on *C* with respect to the QUD in order to account for the mirror image felicity patterns with *only* and *even*. She suggests that the superlative formulation of the scalar presupposition can be used to capture the differences in the types of scales with different operators, which is a question for future research. In this chapter, I focus on the common core of the exclusives, therefore I simply adopt the idea of stronger/weaker alternatives.

In (78) above, I proposed a single lexical entry for *tylko/zaledwie* because *zaledwie* does not block *tylko* from pragmatic scales and its use with numeral scales seems to only be dispreferred, as was shown in example (37) in Section 4.2. However, the approximative and counterfactual uses of *(za)ledwie* require an extra presupposition: the lexical entry for *(za)ledwie_{APPROX}* makes it salient that *not p* could happen, (81c). I observed that those uses require obligatory wide focus (VP or sentence focus), which I take to correlate with the fact that $\neg p$ is a salient alternative. Given that *C* is determined also by the QUD, if $\neg p$ is salient, the context disambiguates between the lexical entries in (78) and (81).

(81) $\llbracket (za)ledwie_{APPROX} \rrbracket^w(C)(p)$ is defined iff

- a. $\exists q \in C[q \geq p \wedge q(w) = 1]$
[Factivity: an alternative at least as strong as *p* is true]
and
- b. $\forall q \in C[q \neq p \rightarrow q > p]$
[Scalarity: *p* is the weakest alternative]
and

- c. $\neg p > p$
 [Scalarity: p is weaker than $\neg p$]

if defined then

$$\llbracket (za)ledwie_{APPROX} \rrbracket^w(C)(p) = \forall q[(q \in C \wedge q \neq p \wedge q > p) \rightarrow q(w) = 0]$$

A separate entry for *dopiero* is needed. This is the only exclusive that is limited to temporal scales and blocks *tylko* and *zaledwie* from temporal uses (in contrast to *only* in English). In Section 4.3 we showed that in contrast to *tylko/zaledwie*, whose scalarity is pragmatically determined, *dopiero* requires a temporal scale. We can build this into its semantics by keeping exactly the same meaning components as for *tylko/zaledwie* in (78) but adding a presupposition specifically about a temporal scale such that the time of p is more noteworthy than the times of the alternatives, either because it was early or because it was late. The exclusive assertion stays the same, because as I noted in Section 4.3, *dopiero* excludes stronger alternatives on the contextual scale, not necessarily later ones (e.g., buying more than 2 tomatoes, (46), vs. finishing earlier than in 2 months, (64a)).⁹

(82) $\llbracket dopiero \rrbracket^w(C)(p)$ is defined iff

- a. $\exists q \in C[q \geq p \wedge q(w) = 1]$
 [Factivity: an alternative at least as strong as p is true]
 and
 b. $\forall q \in C[q \neq p \rightarrow q > p]$
 [Scalarity: p is the weakest alternative]
 and
 c. $\exists t_1 \in T[p = \lambda w[t_1 \text{ in } w]] \wedge \forall q \in C[\exists t_2 \in T[q = \lambda w[t_2 \text{ in } w]] \wedge q \neq p \wedge t_1 >_{\text{NOTEWORTHY}^w} t_2]$
 [Temporality: Time t of p is the most noteworthy.]

if defined then

$$\llbracket dopiero \rrbracket^w(C)(p) = \forall q[(q \in C \wedge q \neq p \wedge q > p) \rightarrow q(w) = 0]$$

As established in Section 4.4, *aż* is the scalar opposite of *tylko/zaledwie*, thus the scale is reverted in the asserted and the presupposed components: (83) vs. (78). The factive presupposition, (83a), states that the prejacent or an alternative at most as strong is true. The scalar presupposition, (83) (ii), is that the prejacent is the highest on the scale. Exclusivity but also scalarity, ‘no lower alternative’, is asserted in the last line of (83).

⁹My intuition is that the exclusion of weaker alternatives should account for why the time of the event in the prejacent is more noteworthy than that of a stronger alternative and then the extra presupposition in (82c) should be superfluous. This remains a topic for future work.

- (83) $\llbracket az \rrbracket^w(C)(p)$ is defined iff
- a. $\exists q \in C[q \leq p \wedge q(w) = 1]$
[Factivity: an alternative at most as strong as p is true]
 - and
 - b. $\forall q \in C[q \neq p \rightarrow p > q]$
[Scalarity: p is the strongest alternative]

if defined then

$$\llbracket az \rrbracket^w(C)(p) = \forall q[(q \in C \wedge q \neq p \wedge p > q) \rightarrow q(w) = 0]$$

Compare now how *az* differs from the additive scalar adverb *even*. The preja-cent presupposition in (84a) expresses additivity, and the scalar presupposition involves a scale of likelihood, (84b) (Guerzoni 2003). There is no contribution to the assertion of the sentence.

- (84) $\llbracket even \rrbracket^w(C)(p)$ is defined iff
- a. $\exists q \in C[q \neq p \wedge q(w) = 1]$
[Factivity and additivity: an alternative q is true]
 - and
 - b. $\forall q \in C[q \neq p \rightarrow q >_{\text{LIKELY}^w} p]$
[Scalarity: p is the least likely alternative]

if defined then

$$\llbracket even \rrbracket^w(C)(p) = p(w)$$

Greenberg (2022) observes that although *even* and *only* are scalar opposites, *even* must place the prejacent high on the scale, “above the norm” (formulated as a universal scalar presupposition in Guerzoni 2003: the least likely among all the alternatives, (84b)). However, *only* “does not inherently express ‘a little’, but just needs to indicate ‘less’ than its alternative(s) in C ” (Greenberg 2022: 23). For this reason *only* is felicitous in the scenario in (85a), but *even* is not felicitous in (85b).

- (85) a. John is tall. He is 1.92m. Bill is shorter – he is only 1.88m (but he is still tall).
Greenberg 2022: (52a) p. 21
- b. John is short. He is 1.52m. Bill is taller. He is (#even) 1.58m (but he is still short).
Greenberg 2022: (52b) p. 21

In contrast to *even/nawet*, *az* is indeed felicitous in the Polish counterpart of Greenberg’s example in (86b), thanks to its contextually defined scale, as opposed to a likelihood scale. (As expected, (86a) shows that *tylko/zaledwie* behave in the exact same way as *only*.)

- (86) a. Jan jest wysoki. Ma 192 cm wzrostu. Bill jest niższy. Ma {tylko / Jan is tall has 192 cm height Bill is shorter has only zaledwie} 188 cm wzrostu (ale i tak jest wysoki.)
merely 188 cm height but even so is tall
'Jan is tall. He is 192 cm. Bill is shorter – he is only 188 cm (but he is still tall).'
- b. Jan jest niski. Ma 152 cm wzrostu. Bill jest wyższy. Ma {aż / #nawet} Jan is short has 152 cm height Bill is taller has AŻ even 158 cm wzrostu (ale i tak jest niski.)
158 cm height but even so is short
'Jan is short. He is 152 cm. Bill is taller. He is aż/(#even) 158 cm (but he is still short).'

Moreover, it can be demonstrated that *aż* cannot use a likelihood scale that matches the wider discourse/world-knowledge. The scale with *aż* must be constructed with reference to the lexical content of the associate, e.g., a non-professional in (87). In (87) *even* adds the reading that hiring a non-professional is very unlikely (under normal circumstances). *Aż* is infelicitous in this sentence because it requires that a non-professional is ranked high on the scale of people we want to hire.

- (87) Potrzebujemy sobowtóra głównego aktora. Zatrudnimy {nawet / #aż}
we.need body.double lead actor will.hire even AŻ
nieprofesjonalistę.
non.professional
'We need a body double of the lead actor. We will even hire a non-professional.'

The factive presupposition of *aż*, (83a), is not additive like the presupposition of *even*, (84a). This means that *aż* can occur in additive contexts, (88a)–(88b), and it can even co-occur with *nawet* 'even', (89).¹⁰

- (88) a. Janek zjadł aż 5 jabłek.
Janek ate AŻ 5 apples
'Janek ate even 5 apples.'

¹⁰I thank the anonymous reviewer for raising this issue.

- b. Janek nie rozmawiał zaledwie z dziekanem, lecz aż z
 Janek not talked merely with dean but AŻ with
 rektorem.
 chancellor
 ‘Janek did not talk merely to the dean but to the chancellor himself.’
 (Tomaszewicz 2012: 325, ex. (7b))

- (89) Maria rozmawiała z szefem zakładu, rozmawiała z dziekanem,
 Maria talked with head institute talked with dean
 rozmawiała nawet aż z rektorem.
 talked even AŻ with chancellor
 ‘Maria talked to the institute head, she talked to the dean, she talked even
 to the chancellor himself.’

The emotive factive test in (90) shows that with *nawet* the assertion is the pre-jacent, but with *aż* it is the excluded component, namely, Maria’s happiness in (90b) is about the fact that she got to talk to somebody high on the scale of important people, not about the fact that she talked to these particular people (the same effect was shown in (66)).

- (90) Maria rozmawiała z szefem zakładu, rozmawiała z dziekanem, ...
 Maria talked with head institute talked with dean
 ‘Maria talked to the institute head, she talked to the dean, ...’
 a. i jest zadowolona, bo rozmawiała nawet z rektorem.
 and is happy because talked even with chancellor
 ‘and she’s happy because she talked even to the chancellor.’
Implied: ‘Maria is happy because she talked to the chancellor.’
 b. i jest zadowolona, bo rozmawiała aż z rektorem.
 and is happy because talked AŻ with chancellor
 ‘and she’s happy because she talked to the chancellor himself.’
Implied: ‘Maria is happy because she talked to somebody no less
 important than the chancellor.’

This also indicates that a shift in the QUD (from ‘Who did Mary talk to?’ to ‘How important a person did Mary talk to?’) changes the set of relevant alternatives in C, pruning some of those included in the prior discourse as shown in Greenberg (2022) for both *even* and *only*. In (88a) *aż* is felicitous although it is true that Janek ate 4/3/2/1 apples.

The formal mechanism of focus association derives the set of alternatives in C (so that the size and location of focus determine the contents of the set) and

this set can be further restricted by the current QUD. Below, I illustrate this for *aż*; the mechanism works exactly the same for any focus-associating item taking sentential scope.

- (91) Condition on focus association (Rooth 1992)

$$C \subseteq \llbracket \alpha \rrbracket^f, \text{ or } C \subseteq \bigcup \llbracket \alpha \rrbracket^f$$

where C is the restrictor of a quantificational adverb and α the sister to $\sim$

In (92) I present the derivation for association with the focus ‘Warsaw’. The meaning of the sentence in (92b) results from the specification of the lexical entry in (83). Note how it does not tell us what the weaker alternatives are. This information is derived from focus association: the focus operator $\sim$ that interprets the feature F on the syntactic constituent is attached at the sentential level, (92a), and thus its restrictor variable S is a subset of the focus alternative value of the TP. The focal presupposition is the set of alternative propositions {‘that Mary drove to Łódź’, ‘that Mary drove to Grójec’, ‘that Mary drove to Berlin’, ...} in (92c). The restrictor of *aż*, C , is a subset of that set satisfying the condition on focus association in (91). Because $q \in C$, the excluded alternatives are a subset of the set in (92c). This subset does not include stronger alternatives such as ‘that Mary drove to Berlin’ because Berlin is further than Warsaw in the context where Mary is driving from Bełchatów (as seen in Figure 1). Obviously, on her way to Warsaw Mary drove through other places, but the QUD is ‘How significantly far did Mary drive?’ so the alternatives, ‘Mary drove to Łódź’, ‘Mary drove to Grójec’ are true, but do not answer the QUD because those places are not significantly far from Bełchatów.

- (92) Mary pojechała aż do [Warszawy]_F.

Mary drove *AŻ* to Warsaw

‘Mary drove all the way to Warsaw.’

- a. $[A\acute{z} \ C] \ [TP_2 \ [\sim \ S] \ [TP_1 \ \text{Mary drove to [Warsaw]}_F]]$
- b. $\llbracket (92) \rrbracket = \lambda w. \forall q [(q \in C \wedge q \neq \llbracket \text{Mary drove to Warsaw} \rrbracket \wedge q < \llbracket \text{Mary drove to Warsaw} \rrbracket \rightarrow \neg q(w))]$
- c. $S \subseteq \llbracket TP_1 \rrbracket^f = \{q : \exists x [q = \lambda w [\text{DRIVE TO}(\text{MARY}, x) \text{ in } w]]\}$
- d. $C \subseteq \llbracket TP_1 \rrbracket^f$ [focus association; see (91)]

When *aż* associates with VP focus, the set C is a subset of the set of different things that Mary did, (93c).

- (93) Mary *aż* [pojechała do Warszawy]_F.
 Mary *Aż* drove to Warsaw
 ‘Mary even drove to Warsaw (of all things she could do).’
- [*Aż* *C*] [_{TP₂} [_~ *S*] [_{TP₁} Mary [drove to Warsaw]_F]]
 - $\llbracket (93) \rrbracket = \lambda w. \forall q [(q \in C \wedge q \neq \llbracket \text{Mary drove to Warsaw} \rrbracket \wedge q < \llbracket \text{Mary drove to Warsaw} \rrbracket) \rightarrow q(w)]$
 - $S \subseteq \llbracket \text{TP}_1 \rrbracket^f = \{q : \exists f \exists x [q = \lambda w [f(\text{MARY}, x) \text{ in } w]]\}$
 - $C \subseteq \llbracket \text{TP}_1 \rrbracket^f$ [focus association; see (91)]

When *aż* associates with sentence focus, the alternatives are all alternative things that could happen, (94c).

- (94) *Aż* [Mary pojechała do Warszawy]_F.
Aż Mary drove to Warsaw
 ‘Of all things that could happen, Mary drove to Warsaw.’
- [*Aż* *C*] [_{TP₂} [_~ *S*] [_{TP₁} [Mary drove to Warsaw]_F]]
 - $\llbracket (94) \rrbracket = \lambda w. \forall q [(q \in C \wedge q \neq \llbracket \text{Mary drove to Warsaw} \rrbracket \wedge q < \llbracket \text{Mary drove to Warsaw} \rrbracket) \rightarrow \neg q(w)]$
 - $S \subseteq \llbracket \text{TP}_1 \rrbracket^f = \{q \in D_{\langle s, t \rangle}\}$
 - $C \subseteq \llbracket \text{TP}_1 \rrbracket^f$ [focus association; see (91)]

Since *aż* may but does not have to have a temporal reading like *dopiero*, I propose that the temporal interpretation is determined by pragmatics and syntax. Namely, TEMPORAL *aż* functions as a complementizer in Polish so it precedes a clause and the focus association mechanism derives a set of alternative propositions which are ordered with respect to time. As noted in footnote 8 the semantics and syntax alone cannot be the reason why *aż* can function as a complementizer but the other exclusive particles cannot. I propose that when *aż* associates with the whole sentence, as in (95b), the set of focus alternatives is the set of answers to the QUD, and pragmatics determines that the temporal ordering is the most relevant. What possibly contributes to this interpretation is the fact that without context a sort of default ordering for multiple events is a temporal ordering. In other words, I assume that, unlike for *dopiero*, for TEMPORAL *aż* no extra presupposition is necessary in the lexical entry. The set *C* in (95d) evokes multiple events and times for the set of propositions derived from wide focus. So what ends up being excluded are alternative propositions describing different events that happened at different times. The scalar presupposition says that those times are ordered

on a scale. The truth conditions say that no weaker proposition, i.e. nothing that happened earlier, is true.¹¹

- (95) Zostaniemy tu aż zajdzie słońce.
 will.stay here AŻ goes.down sun
 ‘We will stay here until the sun goes down.’
- Aż [sun goes down]_F.
 - [Aż C] [TP₂ [~ S] [TP₁ [sun goes down]_F]]
 - $\llbracket (95) \rrbracket = \lambda w. \forall q [(q \in C \wedge q \neq \llbracket \text{sun goes down} \rrbracket \wedge q < \llbracket \text{sun goes down} \rrbracket) \rightarrow \neg q(w)]$
 - $S \subseteq \llbracket \text{TP}_1 \rrbracket^f = \{q : \exists t \in T [q = \lambda w [t \text{ in } w]]\}$
 - $C \subseteq \llbracket \text{TP}_1 \rrbracket^f$ [focus association; see (91)]

Finally, let us see why the four exclusives are interchangeable in the sentence in (1), repeated below in (96). The logical syntax is given in (97a). The focus is on the numeral and the exclusive takes sentential scope. This derives the set of alternatives in (97c). With *tylko/zaledwie/dopiero* the truth conditions say that all of the stronger alternatives are excluded, e.g. ‘that the team achieved 1st place, that the team achieved 15th place, ...’, (97b) and (98b). The scalar presupposition is that the preajcent ‘that the team achieved 30th place’ is the weakest alternative.

- (96) Zespół był słaby i zajął {tylko / zaledwie / dopiero / aż} [30]_F
 team was poor and achieved only merely only_{TEMP} AŻ 30th
 miejsce. (cf. (1))
 place
 ‘The team performed poorly and earned only 30th place.’

¹¹There is an interesting contrast in Polish between two types of *until*-clauses: *aż* and *dopóki*. In contrast to *aż*, (95) repeated as (i.a) below, *dopóki* requires negation for an identical interpretation, (i.b) and my intuition is that this is because no exclusivity is built into its semantics.

- Zostaniemy tu aż zajdzie słońce. = (95)
 will.stay here AŻ goes.down sun
 ‘We will stay here until the sun goes down.’
 - Zostaniemy tu dopóki nie zajdzie słońce.
 will.stay here DOPÓKI not goes.down sun
 ‘We will stay here until the sun goes down.’

Since *aż* excludes lower alternatives on the time scale, we obtain the time point where all alternatives to the sun going down are no longer true. But if *dopóki* works like ‘as long as’, negation is necessary to compositionally derive the endpoint. For extensive work on the Czech counterpart of *dopóki*, *dokud*, see Dočekal (2012, 2015), Iatridou & Zeijlstra (2021).

- (97) a. [TYLKO/ZALEDWIE C] [TP₂ [~ S] [TP₁ the team earned [30th]_F place]]
 b. $\llbracket(97a)\rrbracket = \lambda w. \forall q[(q \in C \wedge q \neq \llbracket\text{the team earned } 30^{\text{th}} \text{ place}\rrbracket \wedge q > \llbracket\text{the team earned } 30^{\text{th}} \text{ place}\rrbracket) \rightarrow \neg q(w)]$
 c. $S \subseteq \llbracket\text{TP}_1\rrbracket^f = \{q : \exists n \in \mathbb{N}[q = \lambda w. \text{THE TEAM EARNED } n^{\text{th}} \text{ PLACE in } w]\}$
 d. $C \subseteq \llbracket\text{TP}_1\rrbracket^f$ [focus association; see (91)]

With *dopiero* the excluded alternatives are further restricted by the presupposition about noteworthiness of alternative times, (98d), e.g., ‘that the team achieved 1st place at time t_1 , that the team achieved 15th place at time t_{15} , ...’. So although the inherent ordering between numerals is such that $30 > 15$, the context defines the scale where achieving 15th is a stronger alternative than achieving 30th place.

- (98) a. [DOPIERO C] [TP₂ [~ S] [TP₁ the team earned [30th]_F place]]
 b. $\llbracket(98a)\rrbracket = \lambda w. \forall q[(q \in C \wedge q \neq \llbracket\text{the team earned } 30^{\text{th}} \text{ place}\rrbracket \wedge q > \llbracket\text{the team earned } 30^{\text{th}} \text{ place}\rrbracket) \rightarrow \neg q(w)]$
 c. $S \subseteq \llbracket\text{TP}_1\rrbracket^f = \{q : \exists n \in \mathbb{N}[q = \lambda w. \text{THE TEAM EARNED } n^{\text{th}} \text{ PLACE in } w]\}$
 d. $C \subseteq \llbracket\text{TP}_1\rrbracket^f$ [focus association; see (91)]
 e. $\exists t_1 \in T[p = \lambda w. \text{THE TEAM EARNED } 30^{\text{th}} \text{ PLACE at } t_1 \text{ in } w] \wedge$
 $\forall q \in C[\exists t_2 \in T \exists n \in \mathbb{N}[q = \lambda w. \text{THE TEAM EARNED } n^{\text{th}} \text{ PLACE at } t_2 \text{ in } w]] \wedge q \neq p \wedge t_1 >_{\text{NOTEWORTHY}^w} t_2]$
 [presupposition in (82c) restricting the set in (98d)]

With *aż* the scalar presupposition is different: the prejacent is the strongest alternative and the ordering in C is ‘that the team achieved 30th place’ $>$ ‘that the team achieved 15th place’. The wider context makes it clear that achieving 30th place is ‘less desirable’, ‘less than expected’ than achieving a place lower on the numeral scale.

- (99) a. [AŻ C] [TP₂ [~ S] [TP₁ the team earned [30th]_F place]]
 b. $\llbracket(99a)\rrbracket = \lambda w. \forall q[(q \in C \wedge q \neq \llbracket\text{the team earned } 30^{\text{th}} \text{ place}\rrbracket \wedge q < \llbracket\text{the team earned } 30^{\text{th}} \text{ place}\rrbracket) \rightarrow \neg q(w)]$
 c. $S \subseteq \llbracket\text{TP}_1\rrbracket^f = \{q : \exists n \in \mathbb{N}[q = \lambda w. \text{THE TEAM EARNED } n^{\text{th}} \text{ PLACE in } w]\}$
 d. $C \subseteq \llbracket\text{TP}_1\rrbracket^f$ [focus association; see (91)]

Thus, the example in (1) is a special case where the truth conditions end up the same with opposite scalar orderings of alternatives, thanks to the context.

6 Family members in other Slavic languages

In Section 3, I discussed the different syntactic configurations for focus association and noted that not all of them may be available in every language; e.g., in English *only* in the clause-initial position cannot associate with the whole clause as it is possible in Polish or German. Now that I identified the minimal semantic differences between the four exclusives in Polish, we can expect that those same parameters will have different settings cross-linguistically and not all of the options may be available or, conversely, there could be more options for variation than in Polish.

For instance, the limits on scale dimension that are lacking with English *only*, are also found with German *erst* (König 1991), a temporal exclusive like *dopiero*. Several other Slavic languages have a counterpart of *až*, *až* in Czech, Slovak and Russian, and *čak* in Bulgarian, Serbian and other South Slavic languages (Tomaszewicz 2013a,b), but they seem to have various requirements on the type of scale. While in Russian *až* is highly expressive and its use is considered colloquial, at first sight, the same range of contexts I have considered here for Polish seems to be available except for the two scope options with negation (Katja Jasinskaja, p.c.). In Czech and Slovak, on the other hand, there seems to be a strong requirement for temporal or distance scales, specifically with DP focus. Thus (100) in Czech does not have the reading that the manager is a noteworthy person for Maria to talk to, but instead a temporal scale needs to be accommodated, as indicated by the translation (Radek Šimík, p.c.)¹². In Polish this reading is obtained with *dopiero* in this sentence.

- (100) Maria mluvila až s manažerem.
 Maria talked AŽ with manager
- a. ‘Maria talked to the manager who appeared late and Maria talked to him but not the people who appeared before him.’
 - b. *Unavailable*: ‘Maria talked to somebody so important as the manager.’ (Czech; Radek Šimík, p.c.)

¹²There may be variation among the speakers. As reported in Tomaszewicz 2013b for the example in (i) some of my informants preferred the scalar interpretation in (a), rather than the temporal interpretation in (b). A dialectal difference is possible (Věra Dvořák, p.c.) – the speakers who preferred (i)a are from Moravia.

- (i) Hanka poprosila o pomoc až prezidenta. (Czech; Tomaszewicz 2013b: (54))
 Hanka asked for help AŽ president
- a. ‘It was nobody less important than the president who Hanka asked for help.’
- b. ‘It was no sooner than when talking to the president that Hanka asked for help.’

With temporal scales *až* in Czech has the same exclusive reading as Polish *zaledwie*, *dopiero* and *až* in the example we saw in (32), i.e., where it can be translated as *only* in English but is incompatible with Polish *tylko*. The sentence in (101) is the counterpart of the Polish (32). Just like in Polish, the reading with *až* is different from the reading with *even*. With *až* the sentence tells us that 6 o'clock is a noteworthy time to wake up, because it is late, i.e., 'no sooner than'. With *even* the meaning is 'Dan woke up several times in addition to the unlikely time at 6 AM'.

- (101) Daneček se vzbudil až v 6 ráno.
 Dan self woke.up AŽ at 6 morning
 a. 'Little Dan woke up only/no sooner than at 6 AM.'
 b. *Unavailable*: 'Little Dan woke up even at 6 AM.'
 (Czech; Tomaszewicz 2013b: (6))

However, *až* in Czech can also associate with a VP and evoke a pragmatic scale (Tomaszewicz 2013a,b). In (102) the alternatives are different things that could result from the boy's surprise; there is no implication of a temporal ordering.

- (102) Chlapec v zeleném overalu se překvapením až [_{VP} začervenál].
 boy in green overalls self surprise AŽ reddened
 'The boy in a green overalls as much as blushed with surprise.'
 (Czech; K. Čapek: *Od člověka k člověku III*)

In Slovak *až* can have temporal readings when modifying a DP with a temporal reading, such as 'Friday' in (103a)–(103b), without the high-on-the-scale implication that is found in such cases in Polish, i.e., in both (103a)–(103b) the meaning is 'until Friday' with the implication that Friday is included ('through Friday' in American English). Compare how the Polish counterpart of (103a) contains a preposition *do* 'to', (104), and in the presence of *až* ('*až do Piątku*') the meaning would be that Friday is late, which would be very odd in this context (we saw this effect in (64c)). The temporal *až* in Slovak is reminiscent of the complementizer *až* in Polish, in that a temporal endpoint is specified without evoking noteworthiness.

- (103) a. V dňoch pondelok až piatok bude predajňa zatvorená.
 in days Monday AŽ Friday AUX store closed
 'In days from Monday to Friday, the store will be closed.'

- b. Keby budeš budúci týždeň na služobke? Utorok až piatok.
when AUX next week on business.trip Tuesday Až Friday
'When will you be on a business trip next week? Tuesday through Friday.'

(Slovak; Andrea Kellö Žáčoková, Jana Bašnáková, p.c.)

- (104) W dniach od poniedziałku do piątku sklep będzie zamknięty.
in days from Monday to Friday store AUX closed
'In days from Monday to Friday, the store will be closed.'

The following example, (105), shows that in Slovak *až* can be used with intervals in general (not just with temporal endpoints as above) and again in Polish in such cases the preposition *do* must be used without *až*, (106), or else the sentence would carry the implication that 400 is significantly more than 384 pages.

- (105) Naša kniha bude mať 384 až 400 strán.
our book AUX have 384 Až 400 pages
'Our book will be 384 to 400 pages long.'

(Slovak; Andrea Kellö Žáčoková, Jana Bašnáková, p.c.)

- (106) Nasza książka będzie mieć od 384 do 400 stron.
our book AUX have from 384 to 400 pages
'Our book will be 384 to 400 pages long.'

The differences in the requirements on the scalar dimension with *až/aż* in Polish, Czech and Slovak predict that in different languages we may find exclusives specified for different scales. With temporal scales either an inherent ordering is lexicalized (Czech and Slovak *až*) or the context specifies the direction of the scale (so that the prejacents is either later or earlier than expected, as with Polish *dopiero*). And while for Polish *dopiero* I opted to assume an extra presupposition to account for temporality, for the Slovak *až* I would consider a meaning without a scalar presupposition but encoding scalarity in the assertion (see Charnavel 2015 for French scalar opposites of *even* and *only*).

In South Slavic languages, the adverbial particle *čak* has the same semantics as *až* (Tomaszewicz 2013a,b), and in Bulgarian, which lacks a counterpart of *dopiero*, *čak* fulfills this role. Just like in Polish, in (107) *čak* contributes the reading that Mary is a noteworthy person for me to talk to. In (108) *čak* would be translated as *dopiero* rather than *až* to yield the meaning that 'now' is a noteworthy time for you to tell me, because it is late, because as we saw in Section 4.3 it is more specific: It calls for a contextually relevant ranking of times.

- (107) Govorix čak s Mary.
 talked čak with Mary
 ‘I talked to somebody so important as Mary.’
 (Bulgarian; Tomaszewicz 2013a: (4), Tomaszewicz 2013b: (1))
- (108) Zašto mi kazvaš čak sega?
 why me tell čak now
 a. ‘Why are you telling me only now?’
 b. *Unavailable*: ‘Why are you telling me even now?’
 (Bulgarian; Tomaszewicz 2013a: (10))

Notably, Bosnian-Croatian-Serbian (BCS) lacks a single lexical item with the meaning of *even* (recall the semantics of *even* in (84), contrasting it with *až* in (83)). Instead, the two meaning components of *even*, scalarity and additivity, are lexicalized separately by *čak* and *i*, where *i* is the conjunction *and* (Jasinskaja & Tomaszewicz 2016). In example (109a) *čak* appears alone, while in (109b) it is accompanied by *i*, which contributes additivity absent in (109a), as evidenced by the continuation.

- (109) a. Poslao sam žalbu čak dekanu, ali nisam nikome drugome.
 sent AUX complaint čak dean but NEG.AUX nobody other
 ‘I sent my complaint to somebody so important as the dean, but I did not send it to anyone else.’
- b. Poslao sam žalbu čak i dekanu, #ali nisam nikome drugome.
 sent AUX complaint čak and dean but NEG.AUX nobody other
 ‘I sent my complaint even to the dean [*Incongruous*:] but I did not send it to anyone else.’ (BCS; Jasinskaja & Tomaszewicz 2016: (27))

In (110a) *čak* together with negation yields the reading ‘low on the scale of distance’, while in (110b) the presence of the additive particle (its negated variant *ni*) yields the ‘low on the scale of likelihood’ reading, which is exactly the reading with *even* in its English translation.

- (110) a. Ivan nije čak u Salzburgu, sad je u Zagrebu.
 Ivan NEG.AUX čak in Salzburg now is in Zagreb
 ‘Ivan is not as far as Salzburg, he is in Zagreb now.’

- b. Ivan nije čak ni u {Zagrebu / #Salzburgu}.
 Ivan NEG.AUX ČAK NEG.and in Zagreb Salzburg
 ‘Ivan is not even in Zagreb / [*Incongruous:*] Salzburg.’
 (BCS; Jasinskaja & Tomaszewicz 2016: (28–29))

The facts from BCS suggest that the different meaning components that I identified for the four Polish exclusives can be lexicalized separately, which adds to the other known cases of compositional combination of particles from the exclusive and additive family, e.g. *auch nur* in German (Guerzoni 2003) (*auch* means ‘also’, *nur* means ‘only’, *auch nur* means ‘even’), *mégis* in Hungarian (Katja Jasinskaja, p.c.; Csirmaz & Slade 2020) (*még* means ‘still’, *is* is an additive particle and *mégis* is a scalar additive particle).

The case of BCS *čak i* shows that the repertoire of exclusive and additive scalar particles uses the same building blocks at the levels of assertion and presupposition.¹³ The presupposed and asserted components may also end up being swapped, as argued in Guerzoni (2003) for subspecies of *even* and *only*, or the strength of particular components may differ, e.g. additivity may be weaker as proposed by Crnić (2011, 2013) for concessive additive particles. Moreover, the differences in presuppositions have grammatical effects, i.e., affect the distribution of different particles in different syntactic environments (Giannakidou 2007).

7 Conclusion

My analysis of the four Polish exclusives *tylko*, *(za)ledwie*, *dopiero* and *aż* adds to the body of work arguing that we can go beyond the identification of the different flavors of exclusive and additive meanings (as in König 1991) but track the origin of those flavors in their minimal building blocks (Rullmann 1997, Guerzoni 2003, Lahiri 2010, Crnić 2011, Coppock & Beaver 2011, 2013, 2014, a.o.). I identified which meaning components are core and thus reoccur in the different exclusive particles and which components require specific tweaking of parameters. Different settings for specific parameters may influence the flavor of exclusivity, e.g., force the temporal interpretation with *dopiero*, but may also extend their uses beyond that of variants of *only*: I showed that in Polish the restrictions on the alternative set with *(za)ledwie* yield approximative/counterfactual uses, and with *aż* they result in the meaning of *until*. This work could also be a starting point for the exploration of other expressions with exclusive meaning, such as gradable adjectives, non-focus associating adverbs, etc., for the identification of parameters

¹³The importance of the distinction between the asserted and presupposed content in the meaning of *even* and *only* goes back to Horn 1969.

other than the semantic ingredients that I isolated here. Coppock & Beaver (2011, 2013, 2014) argue that English adverbial *only* and *just*, and adjectival exclusives *mere*, *sole*, *only*, *single*, *exclusive* vary along two parameters: the ontological type of their arguments and constraints on the question under discussion. Greenberg (2016, 2022) suggests that the variability in the dimensions of the scale could be captured if the scalar presupposition is defined in terms of degrees, with respect to a contextually supplied gradable property. For Polish the next step would be to investigate the building blocks and find a common core with *even*-like particles (*nawet*, which we saw in this chapter, and *choć*(*by*)), additives (*i*, *też*, *także*, *również*, ...), approximatives (the approximative *ledwie* that we saw here belongs to a rich repertoire: *prawie*, *niemal*, *niemalże*, *nieomal*, *omal*, *mało co*,...) and exceptives (*oprócz*, *poza*, *ponadto*, *zresztą*, ...).

Abbreviations

AUX auxiliary
NEG negation

TOP topic particle
Q question particle

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